CSE 429 – Computer Vision and Pattern Recognition

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Adaptive Bilateral Texture Filter for Image Smoothing

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Project Summary

Project Goal

The main objective of the filter is to smooth textures and small details in an image while preserving important structural edges. Traditional smoothing filters such as Gaussian blur and arithmetic mean filters often remove both textures and edges simultaneously, causing loss of image structures. The proposed method addresses this limitation by adaptively adjusting the filtering strength according to the local image structure.

Approach

The pipeline follows the methodology described in the paper.

- **Gradient Map Computation**
- **Four-Directional Structure Map**
- **Adaptive Scale Map**
- **Fourier-Based Bilateral Filtering**

Evaluation Plan

The implementation is evaluated using both qualitative and quantitative analysis.

Qualitative Evaluation

The following visual outputs will be analyzed:Gradient maps, Structure maps, Adaptive scale maps, Edge detection results before and after filtering, Final filtered images

Quantitative Evaluation

Runtime measurements are collected for different adaptive window sizes ($\eta = 8, 12, 16$) to validate the near $O(1)$ computational complexity as in the paper.

The implementation also computes: SSIM, PSNR (dB), BRISQUE, FSIM and Run time in sec.

against traditional smoothing methods such as Gaussian filtering and standard bilateral filtering.